

Fig. 2: An actual-size PCB layout for the sequential tilt-motion lock

the program (sketch) through standard USB port in the computer.

Construction and testing

An actual-size (Arduino shield type), single-side PCB for the sequential tilt-motion lock is shown in Fig. 2 and its component layout in Fig. 3. Assemble

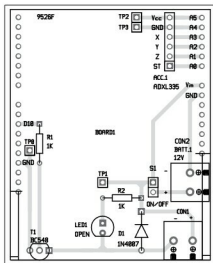


Fig. 3: Component layout for the PCB

the circuit on the recommended PCB to minimize assembly errors.

To test the circuit for proper functioning, switch on S1 and verify correct 12V supply for the circuit at TP1 with respect to TP0.

The neutral position of accelerometer module is parallel to the earth's surface. The +Y and +X axes are

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

marked on the accelerometer module. The default tilt sequence of this circuit defined in the source code is -X, -Y, -X, +Y, -Y. If the tilt sequence is correct, you can observe the glowing of an in-built LED (not shown here) connected to pin 13 of Arduino board. If you want to change the sequence, change it in the source code, recompile the program and burn into the microcontroller.

To open the lock, switch on S1 and tilt the circuit sequentially, as mentioned above, within one minute. If correct action is not completed within one minute, you need to repeat the tilt sequence from the start to unlock it. To close the lock, just switch off the circuit using on/off switch S1. ●

The author is a third-year student of B.Tech (CSE) at the Institute of Engineering and Technology, Alwar (Rajasthan)

To Be Your Reliable Power Partner

MORNSUN® is not only a manufacturer but also provides added value & solutions with our innovative unique designs and products.

Certification: ISO9001:2008, ISO14000, OHSMS18001, and TS16949; UL60950, EN60950, UL60601-1, EN60601-1



DC-DC converters extensive range 1/4W to 100W

AC-DC converters wide range 1W to 120W

Packages Available: PCB MOUNT, CHASSIS MOUNT, SMD, DIP, SIP, and DIN RAIL

MORNSUN®

Email: sales820@mornsun.cn
<http://www.mornsun-power.com>

MORE THAN RELIABILITY

electronica India 2014

Sept. 23-25 BIEC
Booth # 2441